Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 17: Sorting Into Boxes

Equivalence Classes and the Partition Theorem

Chartrand & Zhang, Section 5.2

Name: _____ Date: _____

Two ideas about to meet

In Lecture 14 you met **partitions**: a way of cutting a set into blocks so that every element lands in exactly one. On Wednesday you met **equivalence relations**.
Today's theorem says these are *the same idea*. Every equivalence relation produces a partition, and every partition comes from an equivalence relation.

By the end of today you should be able to

1. Compute the equivalence classes of a relation.
2. State why two classes are always equal or disjoint.
3. Explain the correspondence between equivalence relations and partitions.

1. Equivalence classes

Definition 5.17

Let R be an equivalence relation on a set A. For $a \in A$, the **equivalence class** of a is

$$[a] = \{x \in A : x \, R \, a\}.$$

Think of $[a]$ as the set of all the *relatives* of a.

Since R is reflexive, $a \, R \, a$, so $a \in$ _____. Therefore no equivalence class is _____.

Example 5.18. On $S = \{1, 2, 3, 4, 5, 6\}$, take

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,4), (2,3), (2,6), (3,2), (3,6), (4,1), (6,2), (6,3)\}.$$

$[1] =$ $[2] =$ $[3] =$

$[4] =$ $[5] =$ $[6] =$

Six elements, but only _____ *distinct* classes: _____.

Notice $[1] = [4]$ and $[2] = [3] = [6]$. Different labels, same box.

2. Classes on the integers

For $a \; R \; b$ if $a + b$ is even (Wednesday's relation):

$$[0] = \{x \in \mathbb{Z} : x + 0 \text{ is even}\} = _____$$
$$[1] = \{x \in \mathbb{Z} : x + 1 \text{ is even}\} = _____$$

And $[4] =$ _____, $[-3] =$ _____. Two distinct classes in all.

You have seen this partition before

Evens and odds. That was the first example of a partition in Lecture 14, and it is the partition of $\mathbb{Z}$ produced by this relation.
Try the relation $a \; R \; b$ if $a \equiv b \pmod 3$ and you get _____ classes — the integers leaving remainder 0, 1, or 2.

3. The key theorem

Theorem 5.22 and Corollary 5.24

Let R be an equivalence relation on A, and let $a, b \in A$.

$$\text{If } a \; R \; b \text{ then } [a] = [b]. \qquad \text{Otherwise } [a] \cap [b] = \emptyset.$$

So any two equivalence classes are **either equal or disjoint**. Never partly overlapping.

Theorem 5.23

The distinct equivalence classes of an equivalence relation on A form a **partition** of A.

Why it works — match each partition requirement to the property that supplies it.

Partition requires	Supplied by
Every block is nonempty	
Every element is in some block	
Blocks do not partly overlap	

The overlap argument, in one line

Suppose $x \in [a]$ and $x \in [b]$. Then $x \; R \; a$ and $x \; R \; b$. By _____, $a \; R \; x$; combining with $x \; R \; b$ and using _____ gives $a \; R \; b$, so $[a] = [b]$.
A shared element forces the whole boxes to coincide. **That is why partial overlap cannot happen.**

4. The correspondence runs both ways

Both directions

Relation → partition. The distinct equivalence classes are the blocks. (Theorem 5.23.)
Partition → relation. Given a partition, define $a \; R \; b$ to mean *a and b lie in the same block.* That relation is reflexive, symmetric, and transitive.

Check the second direction. With R meaning "same block":

Reflexive? _____ — a is in the same block as itself.
Symmetric? _____ — same block is a symmetric condition.
Transitive? _____ — if a, b share a block and b, c share a block, it is the same block.

Why this matters

An equivalence relation and a partition are two descriptions of one situation. Whenever you sort objects into groups where every object lands in *exactly one* group — files by extension, integers by remainder, students by section — you have both.
That is why these three properties, and not some other three, are the ones with a name.

5. Going backwards

Let $S = \{a, b, c, d, e\}$ and suppose the distinct equivalence classes are $\{a, c\}$ and $\{b, d, e\}$. Write out R.

Your work

How many pairs? A block of size k contributes _____ pairs, so here $2^2 + 3^2 =$ _____.

Exit ticket

1. For $a \mathrel{R} b$ if $|a| = |b|$ on $\mathbb{Z}$, describe $[3]$ and $[0]$.

2. Can two distinct equivalence classes share exactly one element? Explain.

Before Monday

Read Section 5.3, "Functions," pp. 158–164. A function is a special kind of relation, so this builds directly on today.

Self-check (not collected). Section 5.2, Exercises 5, 9, and 13 — odd-numbered, answers in the back.

Coming up

Homework 6 is due tonight, Friday, September 25, 11:59 p.m.
Quiz 6 is Monday, September 28 — 15 minutes on Sections 5.1 and 5.2.
Homework 7 covers Sections 5.1 and 5.2 and is due Friday, October 2.

Questions I still have